



Hitman is coming to Mumbai

Among the 6 exotic locations used in the upcoming Hitman 2 game, Mumbai will be the 4th.
<https://dgit.in/nov18-17-1>



Wolfenstein 3D made use of Ray Casting algorithms

travel back from an object. A refraction ray could travel through an object but at an angle to mimic the refractive index of the material. Lastly, a shadow ray would be traced towards the light source until it collided with an opaque object. This way, the iterative method of tracing and then breaking down the rays into the three categories allowed for a lot more realism to be added to the rendered image.

SO WHERE DO GPUS COME IN?

The introduction of the highly parallelised GPGPU architecture and the subsequent use of a Unified Shader architecture by both, AMD and NVIDIA, led to a revolution in the computing world. Since the task of tracing each ray is very linear and repetitive, it is the exact same parallelised workload that benefits the most from many parallel cores in your everyday GPU. However, ray tracing across a fixed scene and ray tracing across a dynamic scene with moving objects is a different story altogether. The latter is a lot more intensive owing to the changing nature of the scene. This calls for a lot more parallel cores and improved algorithms that can account

for the dynamic nature of the scene.

Real-time ray tracing was first demonstrated at SIGGRAPH in 2005 using tools developed in 1986 by Mike Muuss. The computation requirement to perform Real-Time Ray Tracing was so much that the REMRT/RT tools developed by Mike relied on a parallel network of

distributed ray-tracing systems. Since 1986, the goal has been to improve the algorithm and to focus on increasing computational power.

NVIDIA's Turing is a milestone in this regard. After achieving the computational power needed to handle fully-fledged VR headsets, the next big thing for GPU manufacturers was to incorporate AI aside from increasing gen-on-gen compute power. There have been several projects that have focused on making real-time ray-tracing happen. Mike Muuss' tools led many others to work on the software. Even today, the original project continues to be in active development as an open source software.

OpenRT, OptiX, OpenRL and many other APIs have now risen to the challenge of enabling real-time ray tracing on GPUs.

RAY TRACING, THE NVIDIA WAY

With the new Turing GPUs, the compute power and the ability to perform real-time ray-tracing is present. However, the algorithms have also increased in complexity to account for greater realism. The number of rays that need to be cast per pixel has now gone up based on the scene as well as the objects that it has to interact with. Thanks to the specialised RT (Ray-Tracing) Cores present in Turing GPUs along with de-noising filtering techniques, this number of rays needed to be cast has been brought down to reasonable levels.



Ray tracing can simulate the effects of a camera



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